Haorui (Harry) Zhang

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PERSONAL PROFILE

I'm Harry Zhang, a PhD student at UW–Madison working on robot learning and physics-based simulation. My research at SBEL has given me a strong foundation in Python/C++ development and digital twins, while my internship with Tesla Optimus brought hands-on experience in GPU-accelerated high fidelity simulation, policy training, and real-robot deployment. I enjoy bringing these pieces together to make robots work reliably beyond simulation. I'm easygoing, enjoy working with others, and take pride in working through challenging problems. Seeing how far AI has come since the early days of ChatGPT, I believe capable, adaptable robots are the next big step. Helping make them useful in everyday life is a journey I'd love to be part of.

INTERNSHIP EXPERIENCE

Tesla, Inc. - Optimus Palo Alto, California
Robot Learning & Simulation Intern *September 2025 - June 2026*

- Developed Tesla's first internal tactile sensor simulation model using hydroelastic contact; validated contact behavior and calibrated the mapping from continuous contact pressure to tactile readings through sim-to-real experiments, enabling planner/controller evaluation and synthetic tactile data generation.
- Increased GPU-batched simulation throughput by 50× over the public solver baseline in large-scale parallel workloads by adding broad-phase collision detection and optimizing hydroelastic-contact collision handling.
- Integrated the new solver, tactile simulation, and whole-body control (WBC) policy replay into the planner team's simulation platform, enabling vision-in-the-loop evaluation and fine-tuning of the combined planner–controller stack.
- Integrated the new physics solver into the controller team's WBC policy training infrastructure, establishing the default workflow for controller training and simulation data collection.
- Deployed and validated WBC policies across physics engines and physical robots, optimizing the data loader, comparing solver variants, and refining armature settings and solver stepping through iterative sim-to-sim and sim-to-real testing. Used observed sim-to-real outcomes to drive targeted improvements to the training infrastructure.

RESEARCH EXPERIENCE

University of Wisconsin-Madison Madison, Wisconsin
Simulation-Based Engineering Lab | Advisor: Dan Negrut *September 2022 - Present*

- Developed a Neural Reduced Dynamics (NRD) framework for distilling high-fidelity physics simulations and enabling high-throughput rollouts for reinforcement learning and differentiable dynamics for gradient-based trajectory optimization.
- Demonstrated the framework across multiple systems with sim-to-sim validation; extending to manipulation and legged locomotion toward a general-purpose learned simulation engine, with sim-to-real transfer validation planned.
- Developed a quadruped reinforcement learning framework in PyChrono, incorporating terrain-aware domain randomization and granular-terrain fine-tuning to improve locomotion performance and generalization.
- Contributed to Project Chrono development and maintenance, supporting its adoption across research projects; helped develop the Autonomy Research Testbed (ART), a ROS 2-based platform enabling a shared autonomy stack to operate in simulation and on physical ground vehicles for sim-to-real experiments.
- Implemented and evaluated ground-robot path-following policies in simulation and physical experiments, investigating simulation fidelity and zero-shot sim-to-real transfer.

RESEARCH EXPERIENCE (CONTINUED)

UW-Madison / Komatsu Research Collaboration — Madison, Wisconsin
Lab Liaison — *October 2023 - October 2024*

- Delivered a tracked-vehicle digital twin and simulation testbed integrating perception, planning, and control for Komatsu engineers.
- Worked with Komatsu Japan engineers on tracked-vehicle design challenges; developed and validated obstacle-avoidance algorithms in high-fidelity simulation.

University of Wisconsin-Madison — Madison, Wisconsin
ARC Lab | Advisor: Xiangru Xu — *December 2020 - September 2022*

- Implemented onboard model predictive control (MPC) by integrating ACADO numerical solvers with the ROS stack on a ground-vehicle platform.
- Developed autonomous-driving control algorithms for the team that placed third at the national eVGrandPrix competition in May 2022.

EDUCATION

University of Wisconsin–Madison — *Sep 2022 – May 2027 (expected)*
Ph.D., Mechanical Engineering — GPA: 3.8/4.0

University of Wisconsin–Madison — *Sep 2022 – May 2024*
M.S., Mechanical Engineering — GPA: 3.8/4.0

University of Wisconsin–Madison — *Sep 2019 – May 2022*
B.S. (Honors), Applied Mathematics, Engineering, and Physics — GPA: 3.9/4.0

SKILLS

Programming: Python, C++, CUDA, MATLAB, Julia, CMake

Machine Learning / AI: PyTorch, TensorFlow, Keras, Transformers, Diffusion Models, Vision-Language-Action (VLA) Models, Reinforcement Learning, Teacher-Student Policy Distillation

Tools and Platforms: Linux, ROS / ROS 2, Docker, MuJoCo, Isaac Lab, Genesis, Blender

SELECTED PUBLICATIONS

Harry Zhang and Dan Negrut. "Learning the Right Abstraction: Neural Reduced Dynamics for Complex Robot Control." *arXiv:2608.19375, 2026.* Under review at *Engineering with Computers*. Paper

Harry Zhang[*], Ganesh Arivoli[*], Huzaifa Unjhawala, Luning Bakke, Radu Serban, and Dan Negrut. "Data-Driven Bulldozer Blade Control for Autonomous Terrain Leveling." *Advanced Robotics Research*, e202500180, 2025. [*]Equal contribution. Paper

Huzaifa Mustafa Unjhawala, Luning Bakke, **Harry Zhang**, Michael Taylor, Ganesh Arivoli, Radu Serban, and Dan Negrut. "A Physics-Based Continuum Model for Versatile, Scalable, and Fast Terramechanics Simulation." *Journal of Terramechanics*, vol. 124, article 101150, 2026. Paper

Aaron Young, Nevindu M. Batagoda, **Harry Zhang**, Akshat Dave, Adithya Pediredla, Dan Negrut, and Ramesh Raskar. "Enhancing Autonomous Navigation by Imaging Hidden Objects Using Single-Photon LiDAR." *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. Paper

Hang Zhang, **Harry Zhang**, Yujie Wang, Zhenhao Zhou, Dan Negrut, and Xiangru Xu. "Shared Control of Teleoperated Vehicles With Delay-Compensated Safety Filtering." *IEEE Conference on Control Technology and Applications (CCTA)*, pp. 192–199, 2025. Paper